From one bit to any polynomial: QSVT spectral filters for track finding in the LHCb VELO toy

Schematic of paper 1 — sections, key equations, figures

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Schematic, 16 September 2026 (section 1 rebuilt on the paper corpus) — target: Communications Physics, about 5,000 words, six figures, Methods, supplement

One sentence. The one-bit quantum filter's single zero removes the isolated false segments and nothing else; with more roots the same primitive removes the coupled ones; the price is depth per call, which a block encoding priced by detector hits converts into qubits.

Source of every figure and number: the thesis chapter draft `QSVT/Paper_planning/qsvt_thesis_chapter_draft/main.tex` (commit `c506a38b`); the figure ledger `QSVT/Efficiency_Frontier/FIGURE_DATA_SOURCES.md`. Chapter sections are cited as [ch. x.y].

1 Introduction

Prose only. Tracking is a linear system; HHL was too deep and too slow to read out; the one-bit filter kept one bit to cut the readout, and pays for it in false segments; a polynomial keeps the efficiency and removes them.

- The problem: at the High-Luminosity LHC the candidate segments grow as T^2, the real ones as T: $n_{\text{seg}} = 4T^2$, $n_{\text{true}} = 4T$.
- Denby–Peterson: a quadratic energy over segments; relaxed to the reals it is a linear system $A\mathbf{x} = \delta\mathbf{1}$.
- Nicotra et al. (HHL): tracking performance of the classical solver, but circuit depth and readout out of reach.
- Chiotopoulos et al. (the one-bit filter): only which segments are on matters; one bit of phase estimation, two calls instead of hundreds, far fewer samples to read out. The price, never measured: the false segments it keeps.
- This paper: the one-bit filter is the degree-one member of the QSVT family; with more roots the same kind of circuit keeps a comparable efficiency and removes the false segments the single zero cannot; and what that costs.

2 Results 1: from hits to a filter

One corpus for the whole paper: $\sigma_{\text{res}} = 5\,\mu m$, $\sigma_{\text{scatt}} = 0.15\,mrad$ (an ideal high-momentum track), 1% of hits lost, cone $0.2\,rad$, $T = 10$ to 1000, ten events per T. Classical and one-bit filter read at the threshold keeping 99% of true segments; the QSVT read at the lowest threshold with false rate at most 1% (its output is nearly binary).

2.1 The toy, its noise and the window

Five planes; three detector effects; the window ε is a quantile of the true kink distribution, so it loses the same tiny fraction of true continuations at any noise.

$$\sigma_p^2 = \sigma_{\text{scatt}}^2 + 6\,(\sigma_{\text{res}}/\Delta z)^2, \qquad \varepsilon = 3\sqrt{2}\,\sigma_p = 1.70 \text{ mrad}, \tag{1}$$

$$P(\theta > \varepsilon) = e^{-9} = 1.2 \times 10^{-4}, \qquad f(\theta) = \begin{cases} 1, & \theta < \varepsilon \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

Measured on the corpus: kink width 0.39 mrad against 0.40 predicted; true continuations outside the window 8.6×10^{-5}; hit residuals 4.8 μm rms; hit loss 0.9 to 1.1% for $T \geq 100$.

2.2 The Hamiltonian and the classical answer

The event is a matrix; it breaks into small blocks, one per connected group of compatible segments; an isolated segment solves to 1/4, a complete track to 4/11 and 5/11, a broken track and a false pair both to 1/3.

$$E(\mathbf{x}) = \tfrac{1}{2}\mathbf{x}^\top A\mathbf{x} - \delta\mathbf{1}^\top\mathbf{x}, \qquad A = (\gamma+\delta)I - C, \qquad A\mathbf{x} = \delta\mathbf{1}, \qquad s = \gamma + \delta = 4. \tag{3}$$

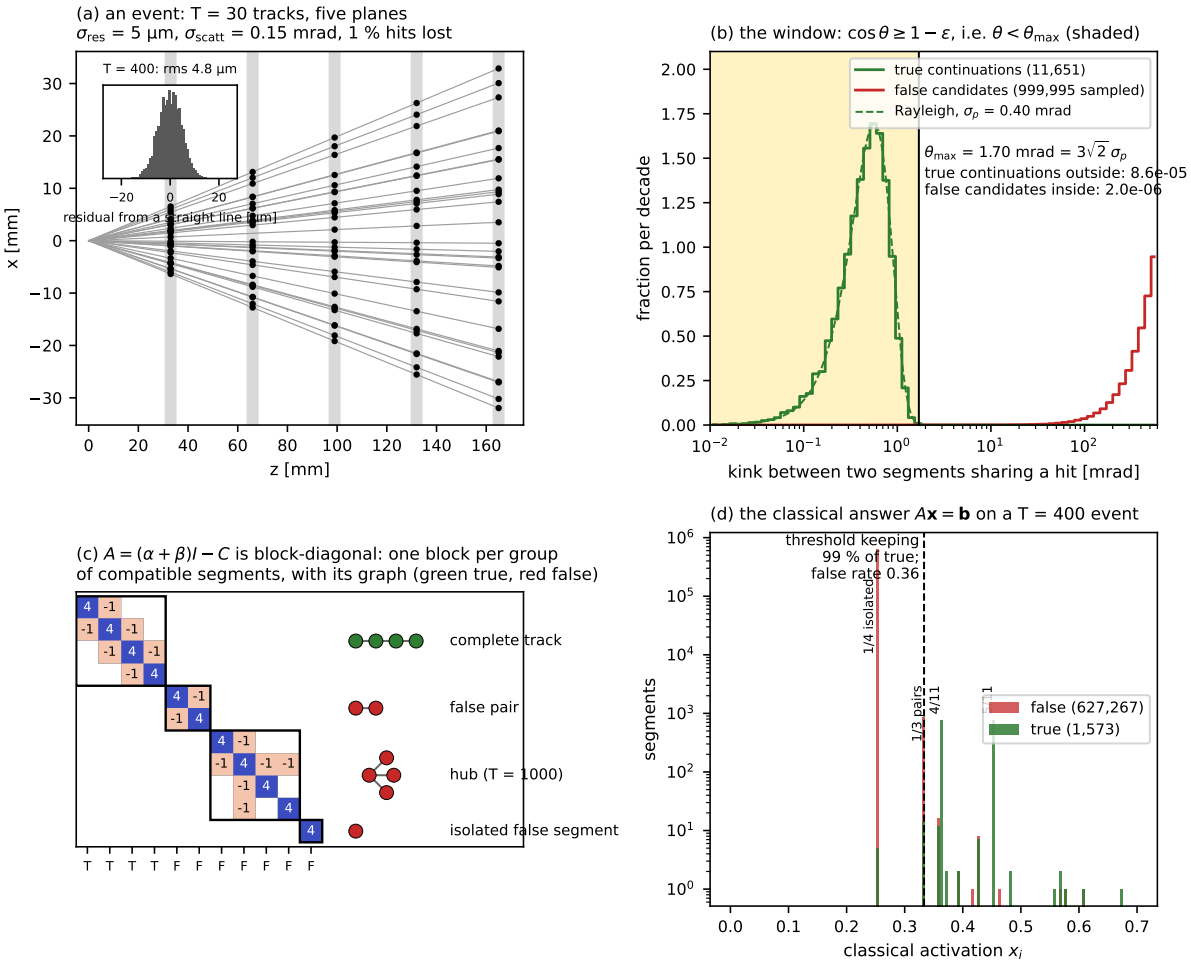


Figure 1: Figure 1. (a) An event and the hit residuals. (b) Kinks of true continuations (Rayleigh, σ_p) and of false candidates, with the window. (c) Real blocks of A from a $T = 400$ event. (d) Classical activations with the 99% threshold. corpus script 10_fig_hits_to_hamiltonian.py

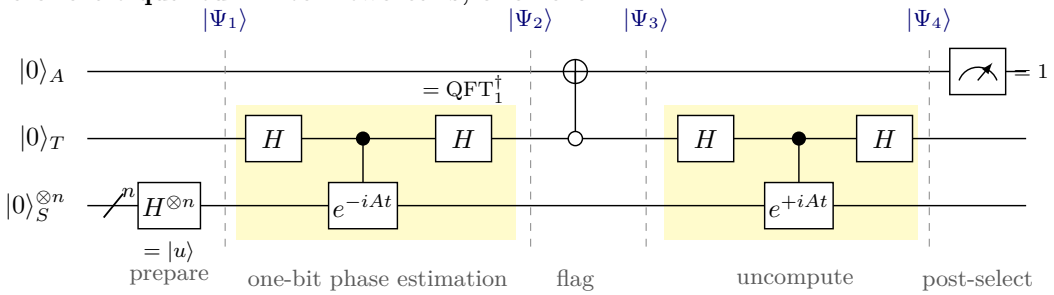
2.3 The one-bit filter, and why it keeps what it keeps

The filter multiplies each eigencomponent by $\cos(\lambda t/2)$*; the zero sits on the isolated line; every other configuration keeps a factor fixed by its eigenvalues, and every hub keeps more of itself than a real track does.*

$$p_i = \Big[\cos^2\big(\tfrac{At}{2}\big)\mathbf{u}\Big]_i^2 + \Big[\tfrac{1}{2}\sin(At)\mathbf{u}\Big]_i^2, \qquad t = \frac{\pi}{s}, \tag{4}$$

$$|\cos(\lambda t/2)|: \ 0 \text{ (isolated)}, \ 0.38 \text{ (pair)}, \ 0.53 \text{ (triple)}, \ 0.59 \text{ (track)}, \ 0.63 \text{ and } 0.77 \text{ (hubs of 3 and 5 prongs)}. \tag{5}$$

(a) the one-bit quantum filter: two calls, one zero



(b) the QSVT: d calls, any bounded polynomial of degree d

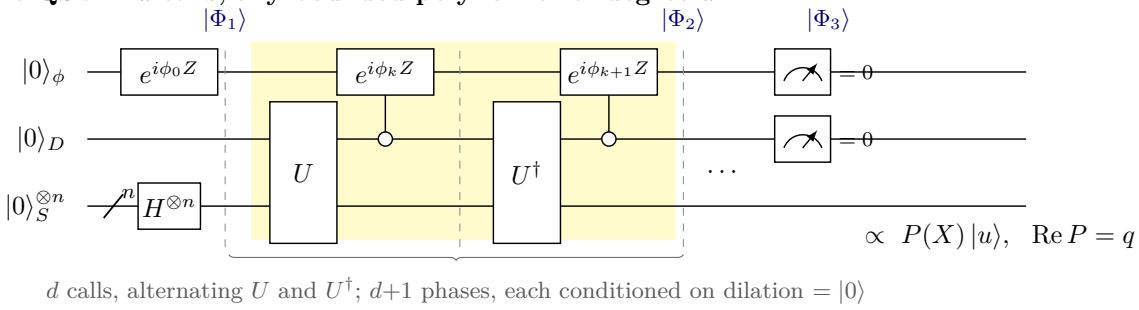


Figure 2: Figure 2. (a) The one-bit filter, correctly wired. (b) The QSVT with the dilation encoding. The difference is boxed. hand-drawn TikZ

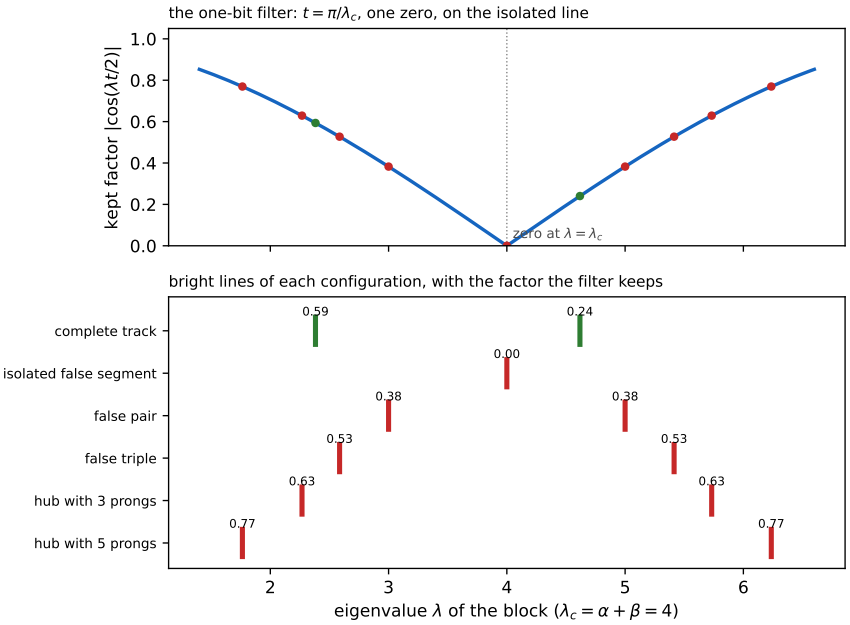


Figure 3: Figure 2 (companion). The kept factor on each configuration's bright lines. corpus script 11_fig_onebit_response.py

2.4 What each solver keeps

Noise creates seven error types. At 99% efficiency the classical solver and the one-bit filter keep every false pair; the QSVT keeps every complete track and removes every false pair; its lost efficiency is broken and contaminated tracks only.

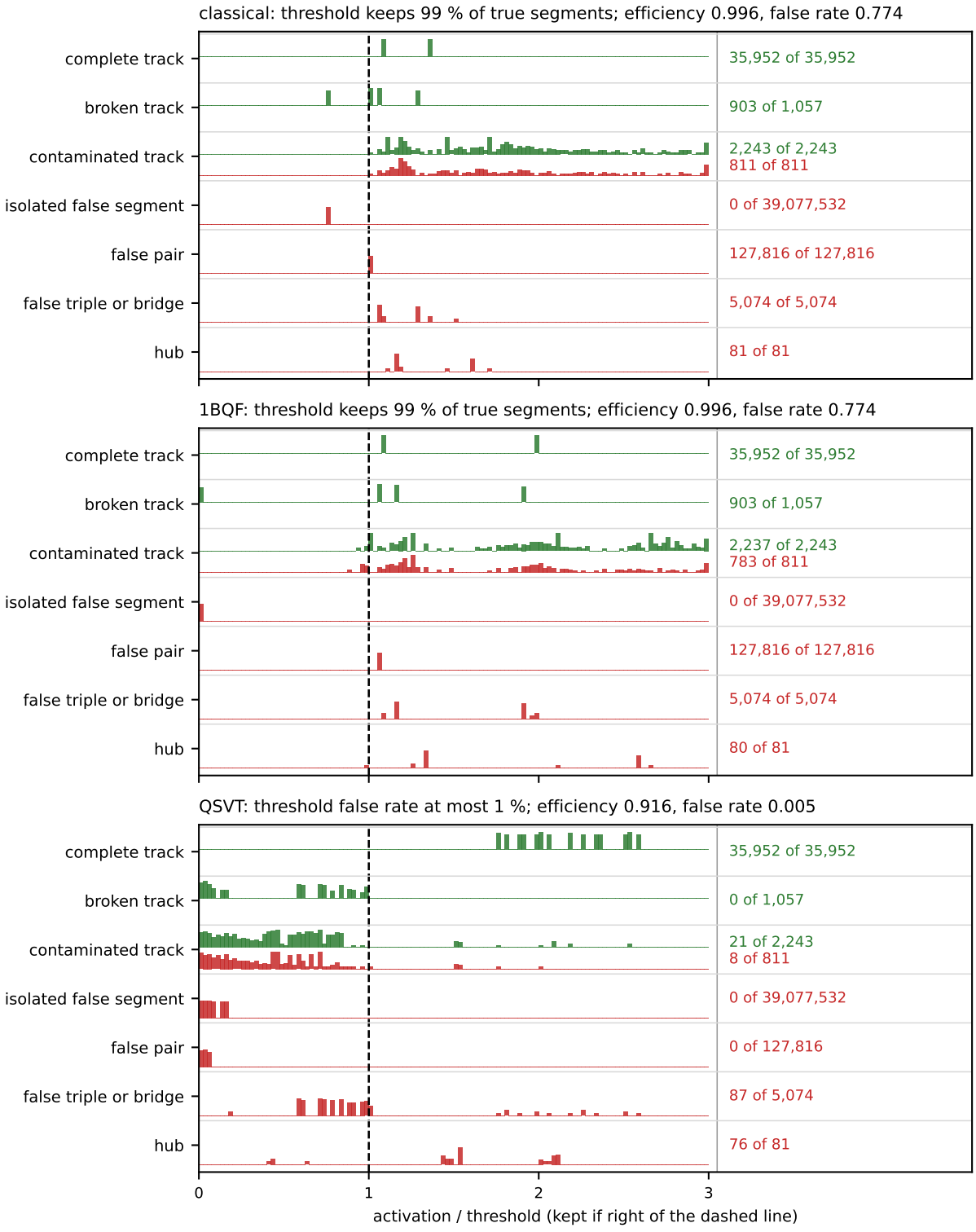


Figure 4: Figure 3. Activation by error type at $T = 1000$, ten events pooled, with each solver's threshold; counts of kept segments at the right of each row. Companion at $T = 400$ in the supplement. corpus script 12_fig_error_atlas.py

	$T = 200$	$T = 400$	$T = 700$	$T = 1000$
classical: false rate at 99% efficiency	0.12	0.35	0.63	0.77
one-bit filter: false rate at 99% efficiency	0.12	0.35	0.63	0.77
QSVT: efficiency at false rate $\leq 1\%$	0.984	0.973	0.937	0.917
QSVT: lowest efficiency over the ten events	0.969	0.938	0.919	0.893

medians over ten events; corpus outputs_5um/summary_by_event.csv

At $T = 1000$ (ten events): the classical solver and the one-bit filter keep all 127,816 false-pair segments and all 5,074 false-triple segments; the QSVT keeps 0 false-pair segments, keeps all 35,952 segments of complete tracks, and loses 1,057 broken-track and 2,222 contaminated-track true segments.

2.5 The one change

Every error type sits on known eigenvalues, so the fix is not a new primitive but more zeros: d calls, any bounded polynomial of degree d (Figure 2b). Results 2 shows where the lines lie; Results 3 which polynomial to apply.

Notes for George (not paper text)

- On most events the one-bit filter and the classical solver read the same false rate at 99%: both thresholds land on the 1/3 level shared by broken tracks and false pairs. The message is "the one-bit filter keeps a high false rate", not "worse than classical".
- Read at the same false rate as the QSVT ($\leq 1\%$), the classical solver keeps 97.5% at $T = 400$ on the median event but only 49% at $T = 700$ and 24% at $T = 1000$; the one-bit filter keeps 49% at all three. The like-for-like advantage is decisive from $T = 700$.
- At $T = 20$ and 50 the 99% reading is volatile (false rate 0 to 0.98 between events): a single broken track sits on the isolated level. Plot medians with the event range.
- The figures were built and every number checked in code; the image viewer was unavailable in this session, so they have not been inspected by eye.

3 The problem as a spectrum

Geometry pins every true track to the same five-hit chain, whose four eigenvalues are known in closed form; every false configuration is a small graph with known lines.

$$\lambda(A) = s - \lambda(C), \qquad \lambda_{\text{true}} \in \{s - 2\cos(k\pi/5)\}_{k=1}^{4} = \{2.382,\ 3.382,\ 4.618,\ 5.618\}, \tag{6}$$

$$x_{\text{outer}}(m) = \tfrac{1}{2} - \frac{r + r^m}{2(1 + r^{m+1})}, \qquad r = 2 - \sqrt{3}. \qquad \text{[ch. 2.4, 2.7]} \tag{7}$$

$$\text{isolated: } \lambda = s; \qquad \text{pair } P_2: \ s \mp 1; \qquad \text{triple } P_3: \ \{s,\ s \mp \sqrt{2}\}; \tag{8}$$

$$\text{hub } K_{1,m}: \ s \pm \sqrt{m} \ \text{ plus } m{-}1 \text{ copies of } s. \qquad \text{[ch. 2.7]} \tag{9}$$

$$\varepsilon = \sqrt{2(\kappa\sigma_{\text{scatt}})^2 + 12\arctan^2(\kappa\sigma_{\text{res}}/\Delta z) + 2\theta_{\text{min}}^2}, \qquad \kappa = 3,\ \Delta z = 33\,\text{mm},\ \theta_{\text{min}} = 15\,\mu\text{rad}. \quad \text{[ch. 2.2]} \tag{10}$$

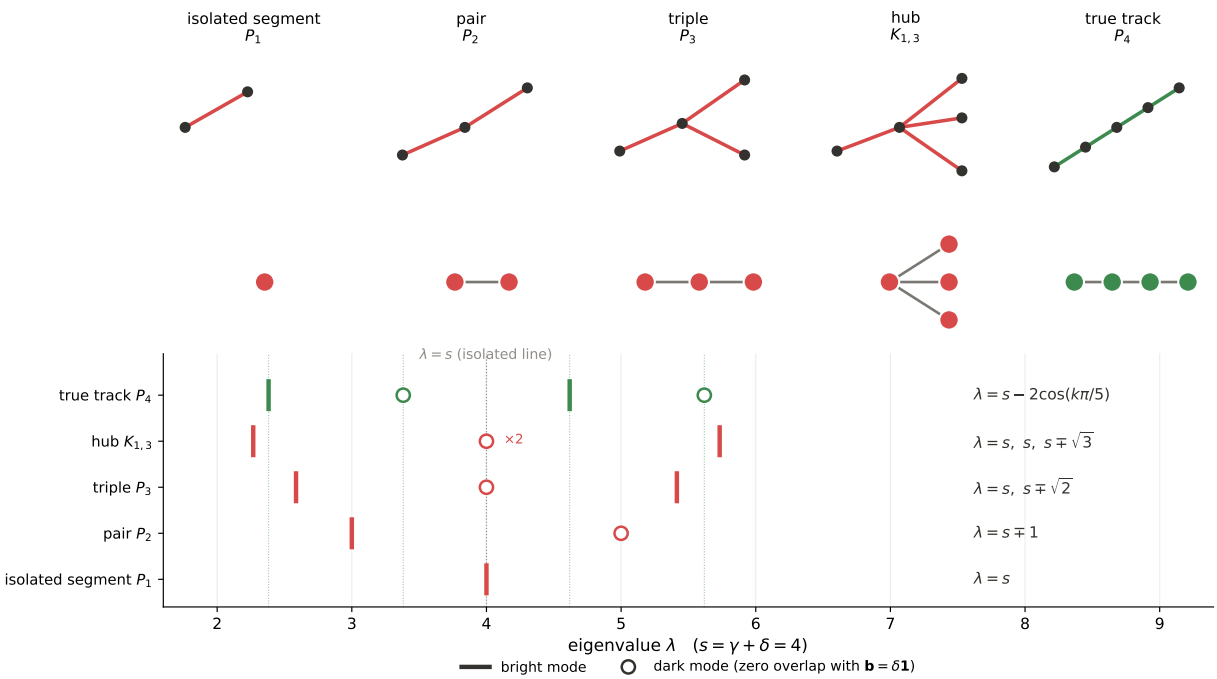


Figure 5: Figure 2. The motif gallery: each cluster type with its eigenvalue lines against the four true lines. fig_motifs, ch. 2.3

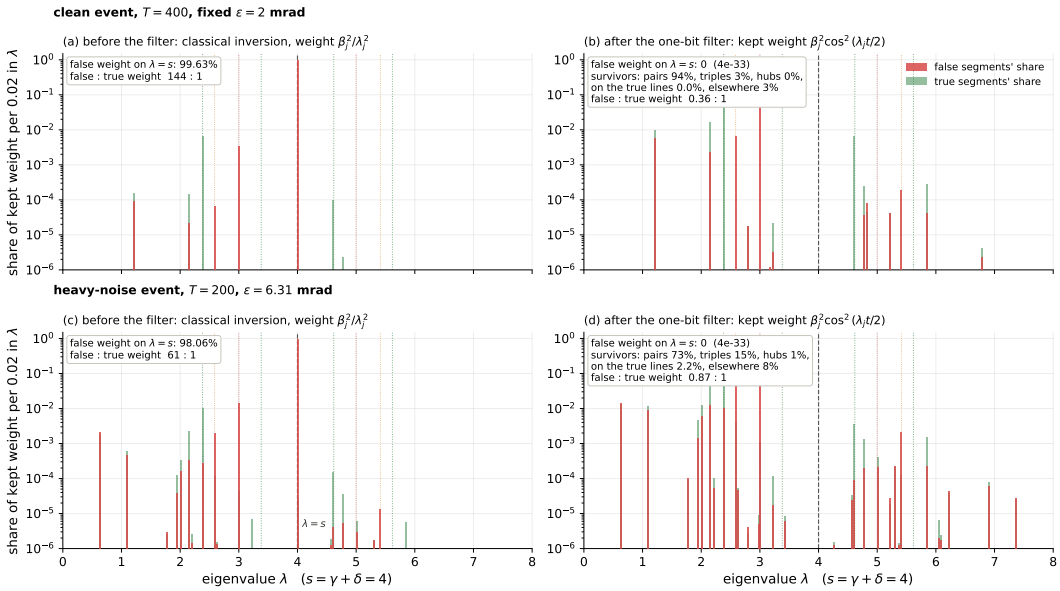


Figure 6: Figure 2 (companion, may go to the supplement). The single zero on the spectrum and where the surviving false segments live. fig_notch_spectrum, ch. 3.2

Numbers: census on clean $T = 200$ events, median true–false gap 0.228, none strictly degenerate with a true chain [ch. 2.7].

4 The QSVT on this primitive

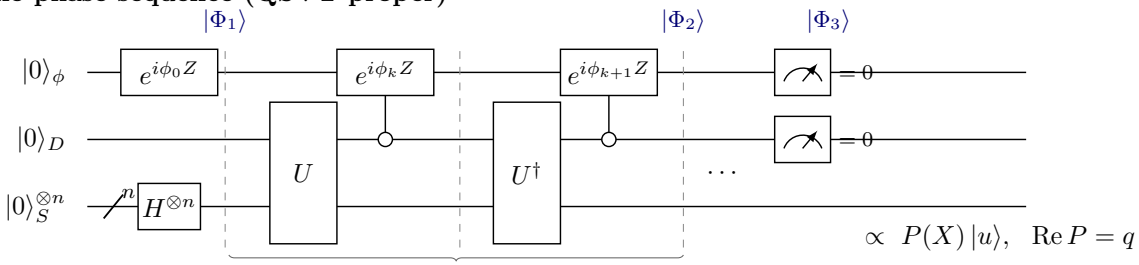
A quantum computer applies any bounded polynomial of A at a cost linear in its degree; the 1BQF is the degree-one member; the comb puts narrow inverse passes on the four true lines and nothing else.

$$(\langle 0| \otimes I)\, U\, (|0\rangle \otimes I) = X, \qquad U_\Phi = e^{i\phi_0(2\Pi - I)} \prod_{k=1}^{d} \Big[U^{(\dagger)} e^{i\phi_k(2\Pi - I)} \Big], \tag{11}$$

$$\langle 0|W^k|0\rangle_{\text{dil}} = T_k(X), \qquad p(X) = \sum_{k=0}^{d} c_k\, T_k(X), \qquad P_{\text{anc}} = \frac{\|p(X)|u\rangle\|^2}{(\sum_k |c_k|)^2}, \tag{12}$$

$$d \;\approx\; \frac{2W}{h} \quad \text{(a feature of width } h \text{ on a spectral span } W\text{).} \qquad \text{[ch. 4.1 to 4.3]} \tag{13}$$

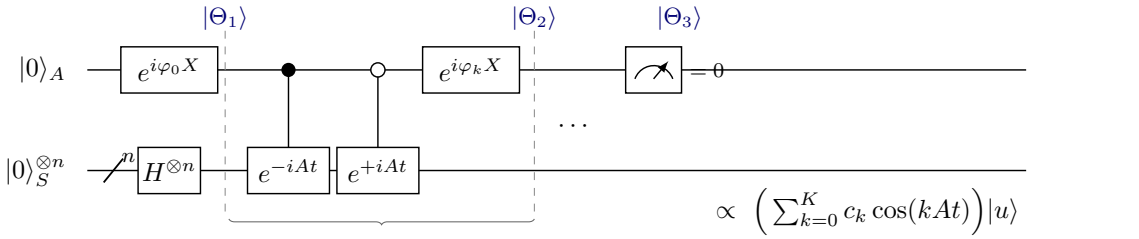
(a) the phase sequence (QSVT proper)



d calls, alternating U and $U^\dagger$; $d{+}1$ phases, each conditioned on dilation $= |0\rangle$

real part: one more ancilla and the sequence with $-\phi$ in superposition; a polynomial without definite parity: one more ancilla, an even and an odd sequence

(b) the trigonometric comb over e^{-iAt} (QETU)



K controlled evolutions, $K{+}1$ rotations $\varphi_0, \ldots, \varphi_K$

the one-bit filter is $K = 1$: $\cos(At/2)$ up to the choice of t; the circuit as built realises e^{-iAt} as a per-pair product (section ch. 11.4)

Figure 7: Figure 3 (candidate). The three realisations of the polynomial: the linear combination of walk powers, the phase sequence, the trigonometric comb over e^{-iAt}. hand-drawn TikZ, ch. 4.2

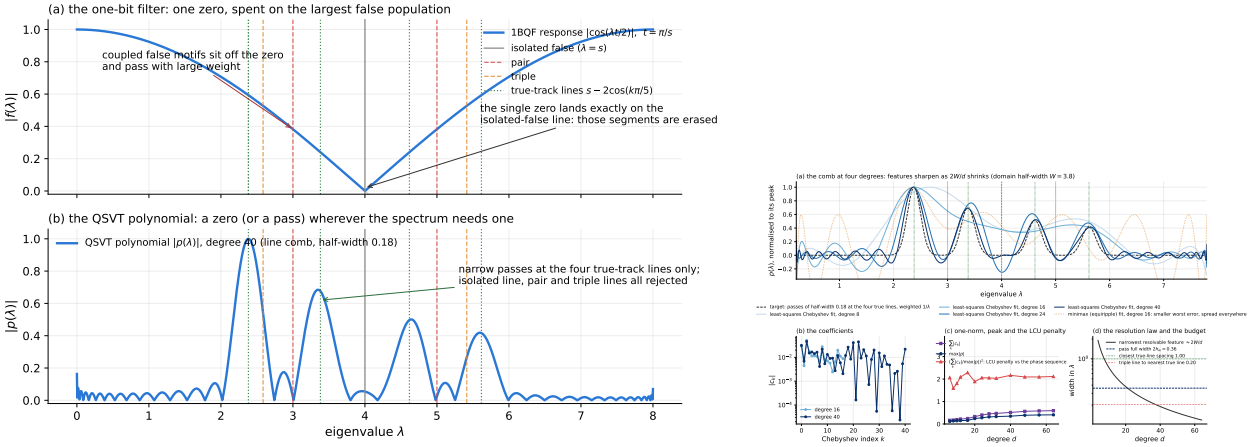


Figure 8: Figure 3. Response functions on the spectrum (the one-bit cosine, the band-limited inverse, the line comb) and the degree ladder. fig_responses, fig_comb_degrees, ch. 4.3

Numbers: production comb degree 40, half-width 0.18; metric-validated degree 12–20 at $T = 200$ [ch. 5.1, 10.1]. The band-limited inverse, the obvious first design, reads a 43.6% false rate at $T = 400$ [ch. 5.1].

5 Results on clean events

Every solver read at the threshold that keeps 99 % of its own true segments; the comb wins through 300 tracks and ties at 400 at the wide 2 mrad acceptance; at the formula acceptance 1000 tracks are solved outright; the limit of every spectral filter is a graph-isomorphism degeneracy.

$$\varepsilon_{\text{seg}} = \frac{n_{\text{true active}}}{n_{\text{true}}}, \qquad f_{\text{seg}} = \frac{n_{\text{false active}}}{n_{\text{active}}}, \qquad \tau: \text{ lowest threshold with } \varepsilon_{\text{seg}} \geq 0.99. \qquad \text{[ch. 2.6]} \tag{14}$$

$$f(PAP^{\top})\,\mathbf{b} = P\,f(A)\,\mathbf{b} \quad \text{for every } f \text{ and every permutation } P \text{ with } P\mathbf{b} = \mathbf{b}. \qquad \text{[ch. 5.5]} \tag{15}$$

The first line is the efficiency-first working point; the second says isomorphic components are twins under every spectral filter, classical scoring included.

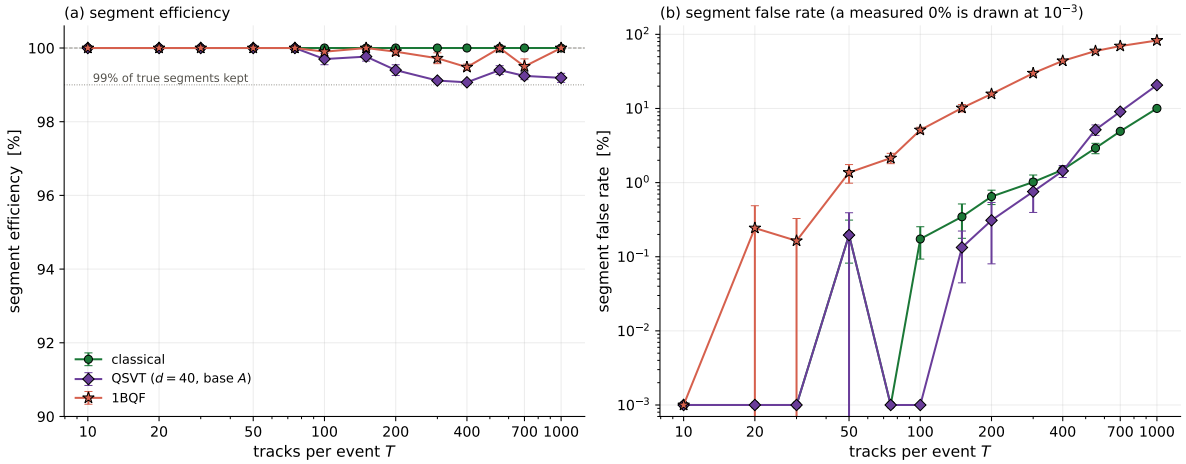


Figure 9: Figure 4. Segment efficiency (a) and false rate (b) against track count for the classical solver, the QSVT (degree 40, base Hamiltonian) and the 1BQF, each at its efficiency-first working point; $\gamma = 3$, 2 mrad, no hit drop. fig_clean2x2, ch. 5.2; the 1BQF curve is the stored package circuit

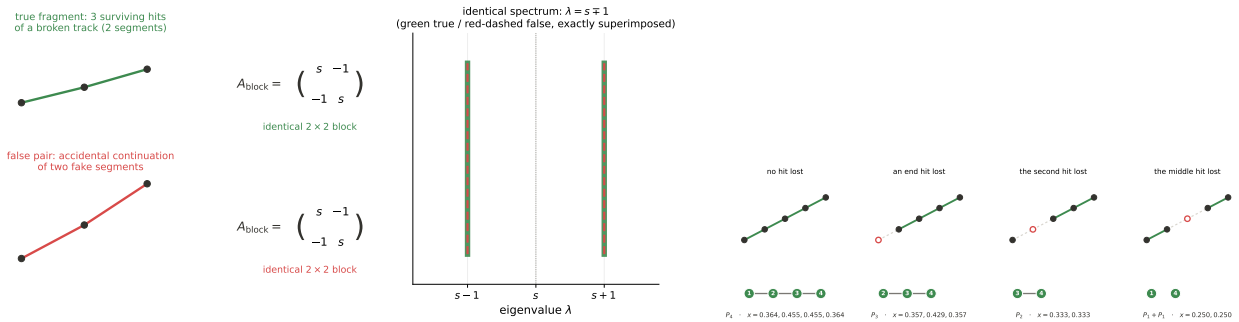


Figure 10: Figure 4 (companions, one paragraph each). Spectral twins; hit inefficiency makes fragments that sit off the comb lines. fig_twin, fig_fragmentation, ch. 5.5, 5.3

False rate at the working point	$T = 200$	$T = 400$	$T = 1000$, formula acceptance
Classical	0.65%	1.49%	—
QSVT, degree 40, base A	0.31%	1.44%	0% measured at $\varepsilon_{\text{seg}} = 99.85\%$
1BQF (package circuit)	—	43.9% (at $\varepsilon_{\text{seg}} = 99.5\%$)	—

ch. 5.2, 10.4; every number at the working point defined above

6 Cost in one currency

All three algorithms call the same controlled evolution; what differs is how many times; width grows as the logarithm of the event, calls are constant in T, and the price is how A enters the circuit: a block encoding priced by hits costs $O(T)$ gates per call instead of $O(T^2)$, for about 45 more qubits.

$$n_{\text{sys}} = \lceil \log_2 4T^2 \rceil + c, \qquad c = 2 \text{ (1BQF, trigonometric comb)},\ 4 \text{ (phase sequence)}, \tag{16}$$

$$c = 7 \text{ (one-norm build)},\ n_c{+}1 \text{ (HHL)}; \qquad 24 \text{ to } 31 \text{ logical qubits at LHCb multiplicity.} \quad \text{[ch. 4.5]} \tag{17}$$

$$\text{calls per shot: } 2,\ 40,\ 63,\ 160,\ 510 \tag{18}$$

$$\text{(1BQF, trigonometric comb, one-norm build, phase sequence, eight-bit HHL)}, \tag{19}$$

$$\text{CX per call: } 152\,T^{1.37}\ (e^{-iAt}, \text{ measured}), \qquad 103\,T^{0.85}\ \text{(hit oracle, modelled)}. \quad \text{[ch. 4.5, 11.5, 11.6]} \tag{20}$$

$$P_{\text{anc}} \approx \frac{0.44}{T}, \qquad \text{repetitions } \frac{1}{P} \to \frac{\pi}{4\sqrt{P}}, \qquad \text{readout } M \ln M,\ M \approx 4T, \tag{21}$$

$$d\,T^{1.5} \log T \text{ walk calls} \quad \text{against} \quad d\,T^2 \text{ classical flops.} \quad \text{[ch. 11.7]} \tag{22}$$

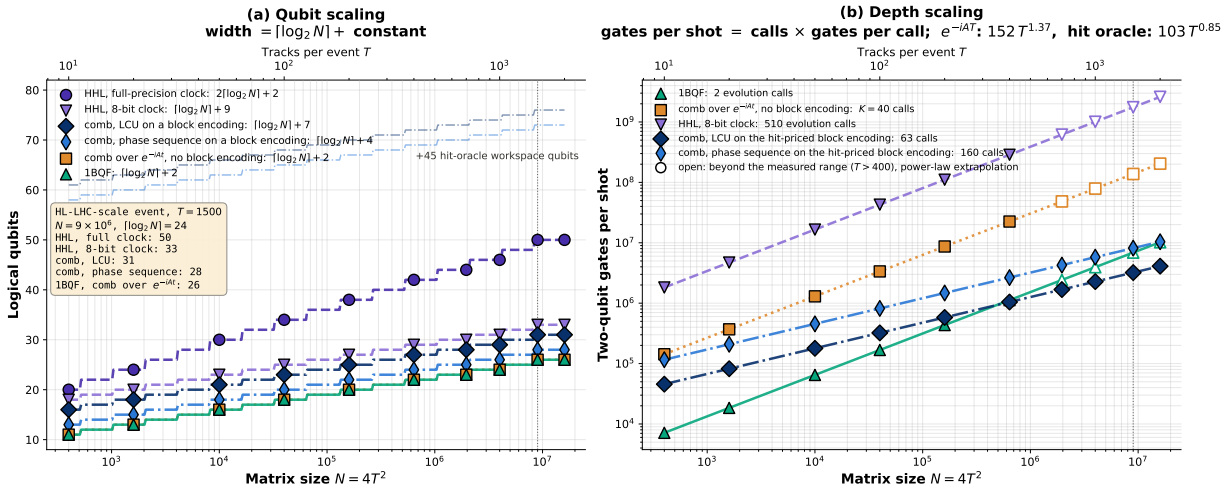


Figure 11: Figure 5. Logical qubits (a) and two-qubit gates per shot (b) against matrix size $N = 4T^2$, in the presentation of the predecessor papers; the HL-LHC-scale event at $T = 1500$ annotated. Transpiled points to be marked against extrapolated ones; the HHL band labelled as an estimate. fig_qubit_scaling, ch. 4.5

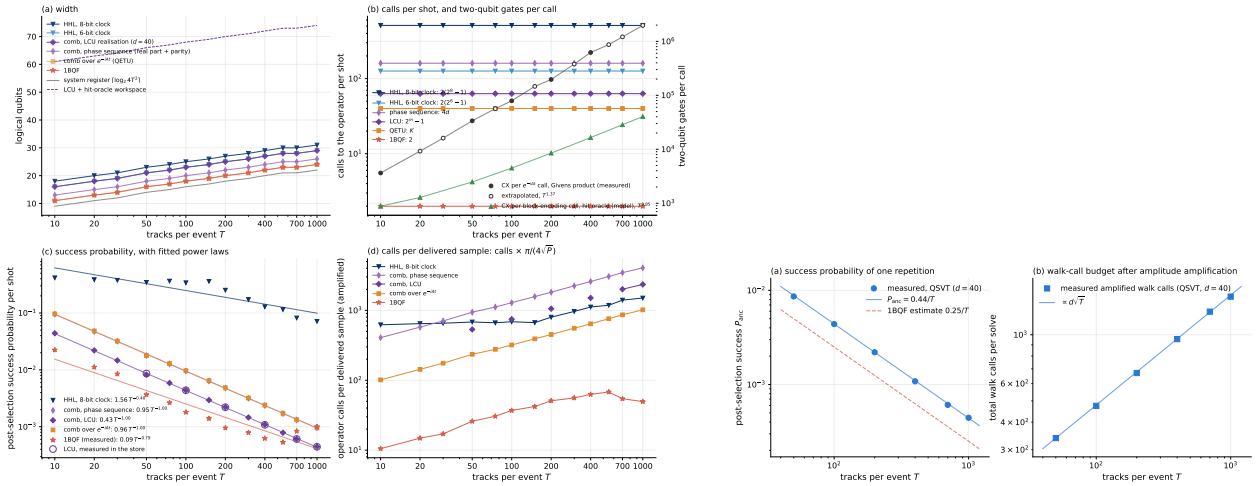


Figure 12: Figure 6. The five-factor resource model (width, calls, gates per call, amplification, readout) for HHL, the 1BQF and the three comb realisations; the measured post-selection probability and the amplified call budget. fig_resource_model, fig_readout, ch. 4.5, 11.7

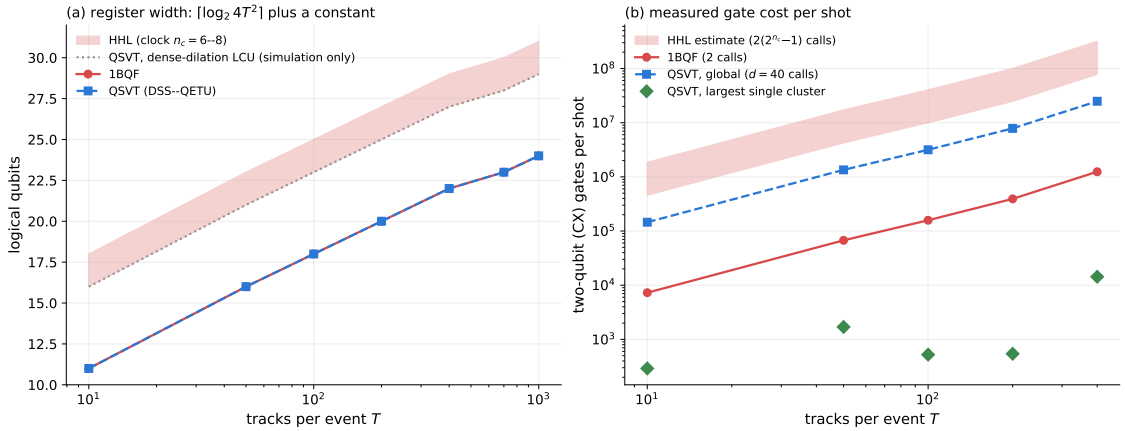


Figure 13: Figure 6 (alternative). HHL, the 1BQF and the comb side by side, global and per-cluster routes, with the surface-code estimate. ch. 11.5; HHL is a structure-level estimate

Numbers: hit oracle 1.6×10^4 CX per call at $T = 400$ against 2.4×10^6 entry-priced; workspace +45–49 qubits over the 21-qubit floor; per-cluster comb about 10^3 physical qubits, global comb about 10^4 [ch. 11.5, 11.6]. Classical loading: hit oracle 0.4 ms and 8 KB at $T = 400$ (paragraph still to be written).

7 Outlook

Noise moves the false weight onto nameable motif lines and demands a refit (next paper); real geometry turns the lines into a band; the per-cluster route is where this meets hardware; nobody has run QSVT on NISQ hardware, so the algorithm is fault-tolerant by construction.

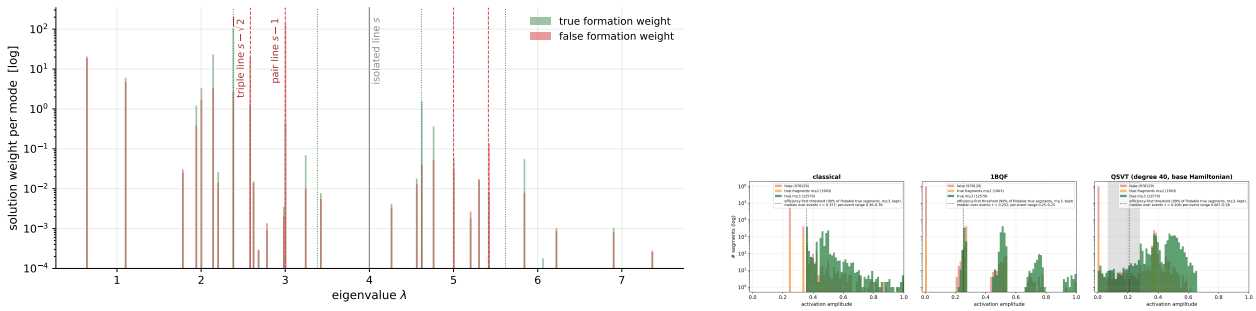


Figure 14: Outlook (optional, one panel at most). The measured false pile-up on the pair and triple lines at heavy noise; activation spectra on real Run-3 geometry, where the true spectrum is a band. ch. 9.1, 11.9

Methods (stub)

Events and the noise triple ($\sigma_{\text{scatt}}, \sigma_{\text{res}}$, hit drop) [ch. 2.1]; the acceptance formula above; segment efficiency, false rate and the efficiency-first working point [ch. 2.6]; the shared data pipeline (events and solutions shared segment for segment across the three solvers); the exact simulation chain (explicit circuit, streaming gate sequence, matrix-free semantics, eigendecomposition agree to 10^{-9}) [ch. 4.4]; which 1BQF object each curve is (exact readout or package circuit).

Supplement map

Paper section	Chapter sections	Figure files	Ledger heading
1 Introduction	1, 3.1, 3.2, 3.6	circuits.tex; amplitude_atlas, chain_amplitude_ladder	fig:circuits; fig:atlas, fig:ladder
2 Spectrum	2.2, 2.3, 2.4, 2.7	motif_gallery; notch_spectrum	fig_motifs; fig_notch_spectrum
3 QSVT	4.1, 4.2, 4.3, 5.1	circuits_qsvt.tex; response_functions, comb_degrees	fig:circuitsqsvt; fig_responses, fig_comb_degrees
4 Results	2.6, 5.2, 5.3, 5.5, 10.4	clean_headline; spectral_twins, fragmentation	fig_clean2x2; fig_twin, fig_fragmentation
5 Cost	4.5, 11.5, 11.6, 11.7	qubit_scaling; resource_model, readout_probability, resource_comparison	fig_qubit_scaling; fig_resource_model, fig_readout; resource_comparison entry
6 Outlook	6, 11.2, 11.9	measured_pileup; run3_activation_spectra	(ledger entries of the same names)

Held for later papers: the heavy-noise census, the fork and occupancy terms, the refit, the degree and density scans, the transfer problem (paper 3); the block-encoding construction (paper 4); the spectral theorems in proof form (paper 2).